

1. Hypergraph Disordered Spin Glass

$$E = \sum_{e \in \mathcal{E}} J_e \prod_{v \in e} \sigma_v$$

Spins: $\sigma_v \in \{-1, 1\}$
Order $\in \{3, 4, 6\}$ (Triangles, Squares, Hexagons)
Couplings $\sim \mathcal{N}(0, 1)$ or ± 1

Rugged energy landscape with exponential metastable trap states

2. Single-Layer Autoregressive Model (SLAS)

$$q(\sigma) = \prod_{v=1}^N q(\sigma_v \mid \sigma_{<v})$$

$$\theta(\sigma_v \mid \sigma_{<v}) = \sigma_v \left(2 \sum_{u < v} W_{vu} \sigma_u \right)$$

Strictly Lower-Triangular $\in \mathbb{R}^{(N \times N)}$

Exact likelihood | Single matrix-mult $((N^2))$
Physical reading: learned pairwise effective coupling

3. Variance-Free Annealing Loop

$$\text{Cooling: } \beta_t = \beta_{\min} + \frac{t}{N_T} (\beta_{\max} - \beta_{\min})$$

Metropolis Buffer: $\mathcal{B} = \{\sigma^{(m)}\}_{m=1}^M$ ($M=10$ local sweeps at inverse temp β_t)

Global Proposal: $\sigma' \sim q_\theta$ with MH test:

$$q(\sigma \rightarrow \sigma') = \min\left(1, e^{-\beta \Delta H} \frac{q(\sigma)}{q(\sigma')}\right)$$

Supervised ML: $\mathcal{L}_{ML} = -\frac{1}{M} \sum \log q_\theta(\sigma^{(m)})$
Zero score-function gradient variance!